EPISODE 6 RECAP

NORMAL/GAUSSIAN RANDOM VARIABLE (model)

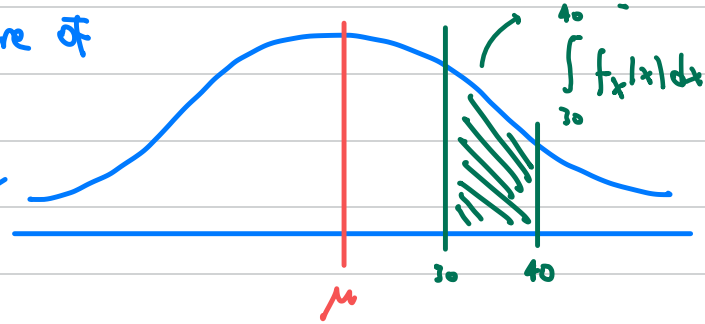
When we write $X \sim N(\mu, \sigma^2)$, X is a NORMAL/GAUSSIAN random variable. $x \in \mathbb{R}$ possible outcomes of X.

Its distribution is defined by two parameters:

μ controls the centre of the distribution

σ controls its dispersion

$$P(30 < X < 40) = \int_{30}^{40} f_X(x)\,dx$$



pdf $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$ → don't need to know

the probability density function is associated to all (absolutely) continuous random variables.

- most popular (VIP) distribution of all, thanks to the CENTRAL LIMIT THEOREM, STAY TUNED!
- used to model measurements error and white noise
- In R to simulate from $X \sim N(\mu, \sigma^2)$ n times: rnorm(n, μ, σ). Also:

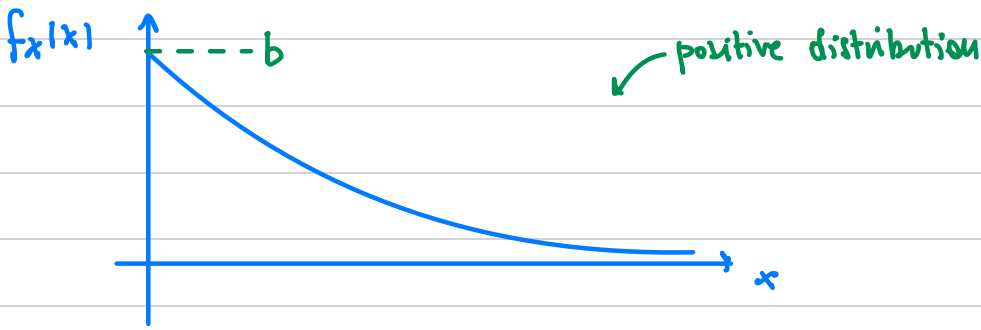
  $P(X \leq x) =$ pnorm(x, μ, σ)

  $P(X \leq q) = \alpha \longrightarrow q =$ qnorm(α, μ, σ)

THE EXPONENTIAL RANDOM VARIABLE (model)

When we write $X \sim Exp(b)$, X is an EXPONENTIAL random variable. $x \geq 0$ possible outcomes of X.

Its distribution is defined by the parameter b



pdf $f_X(x) = be^{-bx}$

the probability density function is associated to all (absolutely) continuous random variables.

cdf $F_X(x) = P(X \leq x) = 1 - e^{-bx}$ survival function $G(x) = P(X > x) = e^{-bx}$

$\hookrightarrow \int_0^x be^{-bt}\,dt$

- simpler distribution for WAITING TIME (in particular if memoriless property is required)
- In R to simulate from $X \sim Exp(b)$ n times: rexp(n,b). Also:

  $P(X \leq x) =$ pexp(x,b)

  $P(X \leq q) = \alpha \longrightarrow q =$ qexp(α,b)